Reg. No. | | | | | | | | | | |

[18MT54]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

PHASE TRANSFORMATIONS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. List out the limitations and applications of phase diagram. 06
   b. What is solid solution and its types? How solid solubility can be predicted using Hume-Ruthery conditions. 08

2. a. Describe the process of constructing isomorphous phase diagram. 07
   b. What are the allotropic transformations observed in iron? Draw the Fe-Fe_3C diagram. Identify the phases and transformation temperatures. 07

3. a. Explain the process of homogeneous nucleation with neat sketch. 08
   b. Draw the free energy change associated with homogeneous nucleation of a sphere of radius 'r'. 06

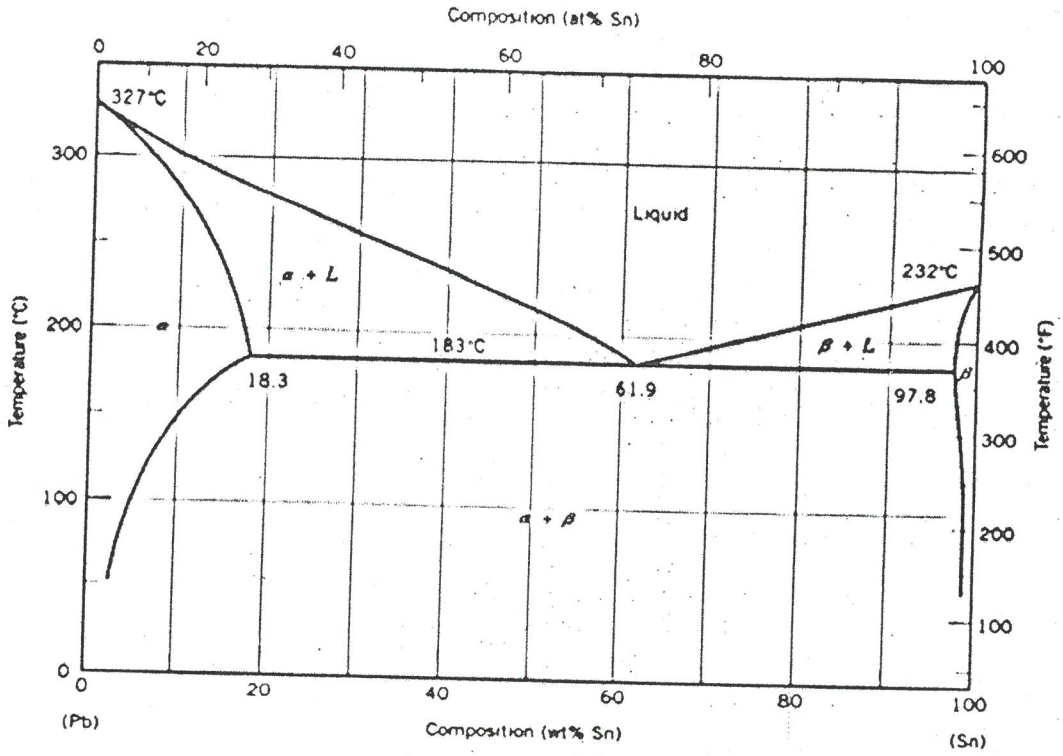
OR

4. a. Explain the process of heterogeneous nucleation with neat sketch. 07
   b. Briefly discuss grain growth, recovery and recrystallization. 07

5. a. Elucidate the homogeneous nucleation of precipitation with suitable sketch. 10
   b. Explain continuous mode of precipitation with suitable sketch. 04

OR

6. a. A 1.5 kg sample of 90 Wt% Pb – 10 Wt% Sn alloy is heated to 250°C; at this temperature it is an entirely α-phase solid solution. The alloy is to melted to the extent that 50% of the specimen is liquid, but the remainder being the α-phase. This may be accomplished either by heating the alloy or changing its composition which holding its temperature constant. 08
   i. To what temperature must the specimen be heated?
   ii. How much tin must be added to the 1.5 kg specimen at 250°C to achieve this state?



b. Explain coherent and semi-coherent precipitates with neat sketch. **06**

7. a. What is Ms Temperature? Explain the effect of Carbon on Ms Temperature. **07**

b. Explain the crystal geometry of martensitic transformation in steel using Bain distortion model. **07**

OR

8. a. Elucidate the stability of austenite during martensitic transformation. **07**

b. Explain athermal transformation characteristics during martensitic transformation. **07**